Ex no: 2

Date: 30/08/23

Study of Different Types of Network cables

Aim:

To study different types of network cables and understand their applications.

a) Understand different types of network cable.

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantage	Application/ Use
UTP	Category 3	10 bps	Advantages * Cheaper in cost * Easy to install Disadvantage * More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet ~~Fast Ethernet~~ GN
	Category 5	upto 100 Mbps		
	category5e	1 Gbps		
STP	category 6, 6a	10 Gbps	Advantages * Shielded * Faster than UTP Disadvantage * Expensive * Or scaler installation effort	Gigabit Ethernet, 10 G Ethernet (55m) widely used in data centres
SSTP	category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100m)
Coaxial cable.	RG-6 RG-59 RG-11	10-100 Mbps	Advantage * High bandwidth * Immune to interference * Versatile Disadvantage * Limited distance * cost * Size is bulky	Speed of signal is 500m relevant network high speed internet connections

Fibre optics cable.	Single mode multimode	100 Gbps	Advantages * High speed * High bandwidth * High security Disadvantages * Expensive * Requires skilled Installers	* Maximum distance of fibre optics cable is around 100 meters

b) Make Your Own Ethernet Cross-Over Cable / Straight cable.

Straight-Through Cable

It uses the same wiring standard (T568A-T568A or T568B-T568B) at both ends.

Used for Connecting:

* PC to Switch
* PC to Router
* Switch to Router

Cross-Over Cable

It uses T568A at one end and T568B at the other end.

Used for connecting

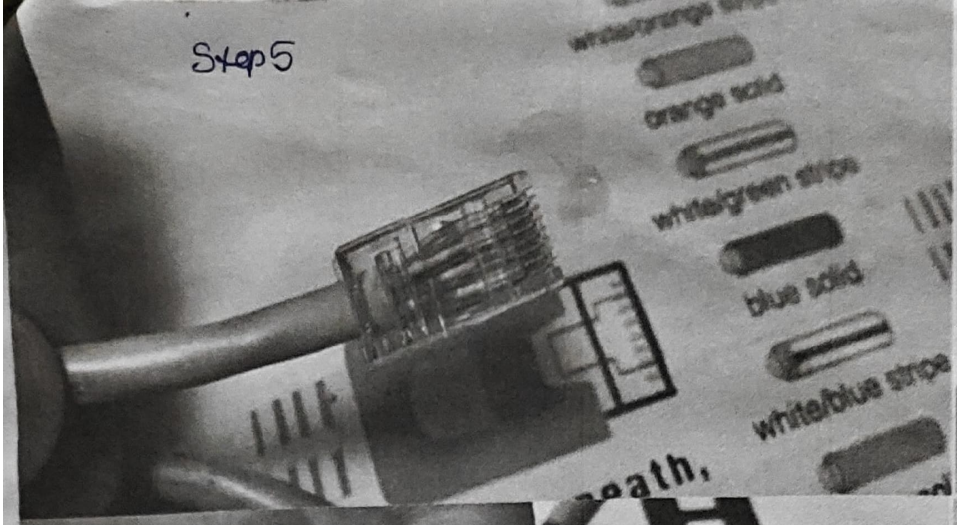
* PC to PC
* Switch to Switch
* Hub to Hub

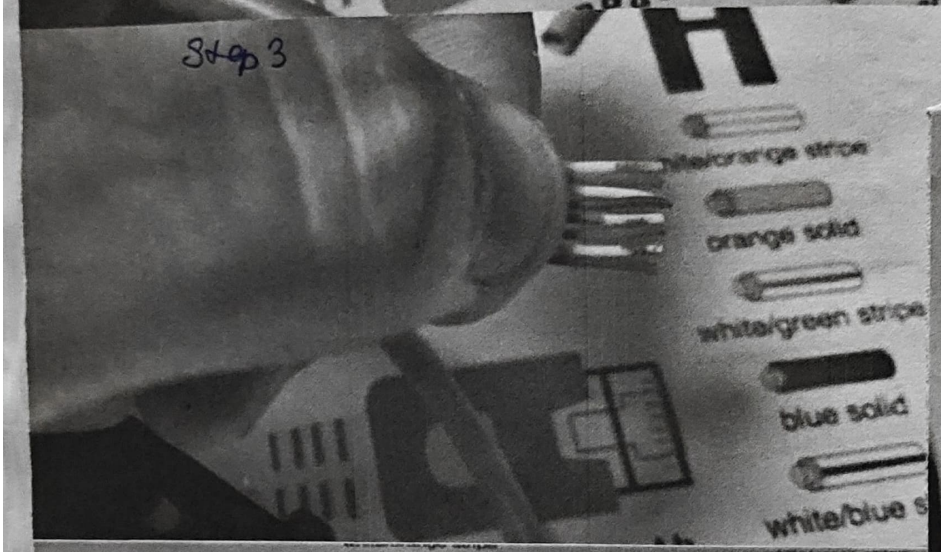
Procedure:

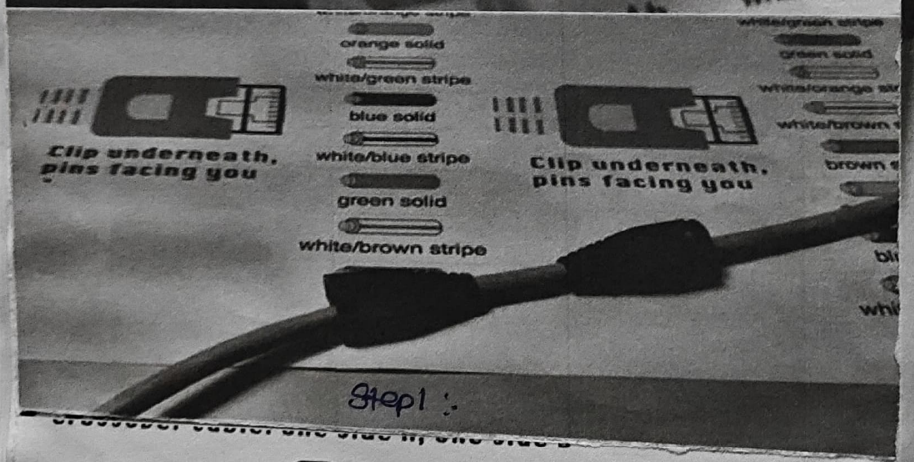
1. To start construction of the device, begin by threading shields onto the cable.
2. Next, strip approx 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.
3. After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B
4. Trim the wires evenly and insert them into the RJ45 connector.
5. Push the cable fully into the connector.
6. Crimp the RJ45 connector using the crimping tool.
7. Repeat for the other end and test the cable.

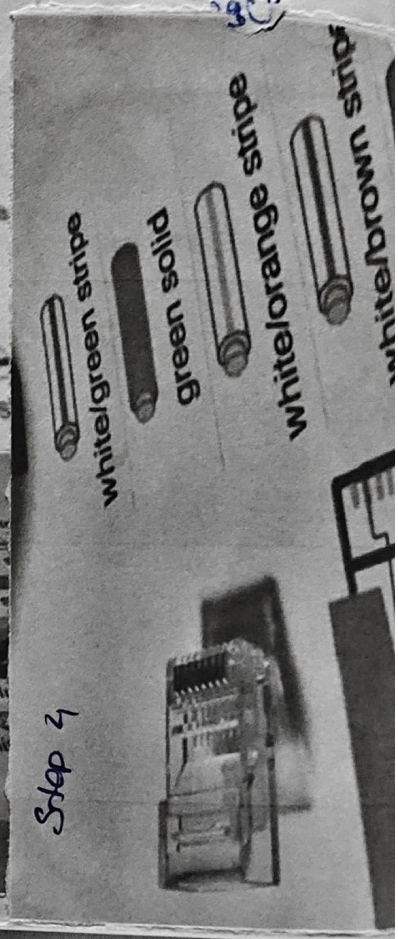
Result:

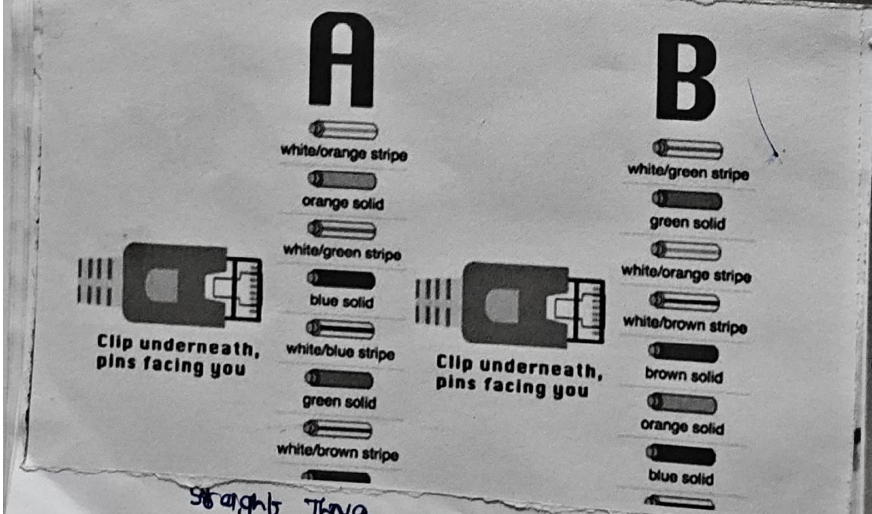
Thus the different types of network cables were studied and straight wire Ethernet cable was prepared and tested successfully and verify

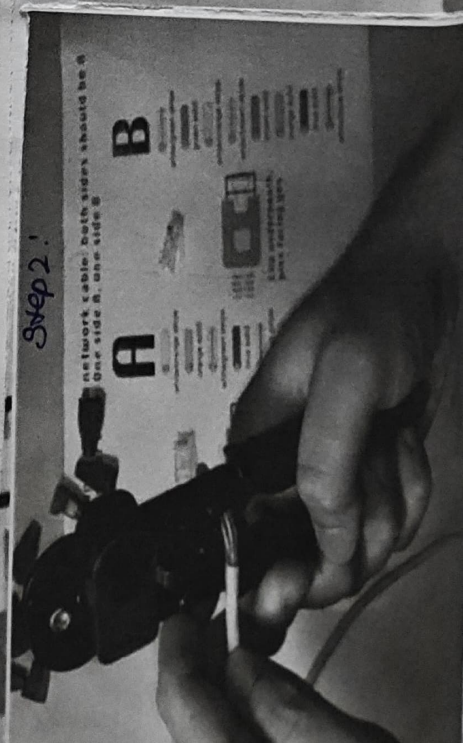

Step 5
orange solid
white/green stripe
blue solid
white/blue stripe

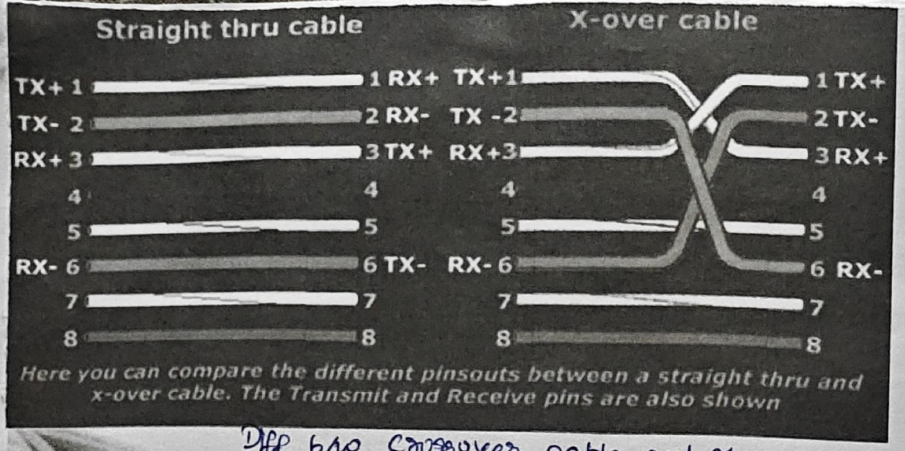

Step 3
orange solid
white/green stripe
blue solid


Clip underneath, pins facing you
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Step 1 :


white/green stripe
green solid
white/orange stripe
Step 4


A
B
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you
white/green stripe
green solid
white/orange stripe
white/brown stripe
brown solid
orange solid
blue solid
Straight Thru network cable.


Step 2 :
A
B



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Diff b/n crossover cable and straight cable

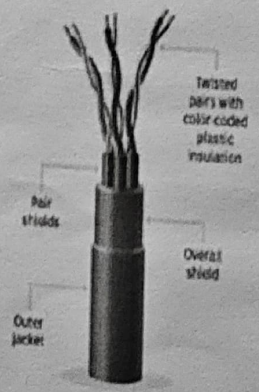


Fibre optics cable

UTP



STP



Shielded twisted pair

Coaxial cable

SSTP

